

DTSA 5002

Statistical Estimation for Data Science and AI

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – Statistics Pathway	AI Engineer (28.2 / 100)	25.8 / 100
SPECIALIZATION Data Science Foundations: Statistical Inference Pathway		

26

CURRENT AVG SCORE
across 6 job roles

58

POTENTIAL SCORE
with recommended additions

+42

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduction to statistical inference, sampling distributions, and confidence intervals. Learn how to define and construct good estimators, method of moments estimation, maximum likelihood estimation, and methods of constructing confidence intervals that will extend to more general settings.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	28.2	57.6	+29.4	MEDIUM
Machine Learning Engineer	28.1	70.5	+42.5	HIGH
Data Scientist	27.9	57.2	+29.2	MEDIUM
Applied Scientist	25.7	55.7	+30.0	HIGH
MLOps Engineer	22.7	64.3	+41.6	HIGH
ML Engineer	22.5	45.2	+22.7	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1
- HIGH
- Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

AI Engineer28.2 → 57.6 (+29.4 pts)

followingdatamodelslearningsoftwaremodelsystemsproblems

Machine Learning Engineer28.1 → 70.5 (+42.5 pts)

datasetslearningtoolsexploratorysciencemodelsmodel

Data Scientist27.9 → 57.2 (+29.2 pts)

datasetsproblems

Applied Scientist25.7 → 55.7 (+30.0 pts)

computationalproblemsmodelsdatalearningsystemsindustryfundamental

MLOps Engineer22.7 → 64.3 (+41.6 pts)

datamodelscientistsystemslearningmachinetoolsmodels

ML Engineer22.5 → 45.2 (+22.7 pts)

datalearningmachinesystemsmodelmodelworldproblems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.